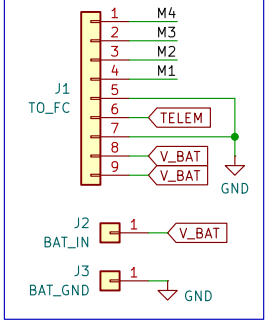
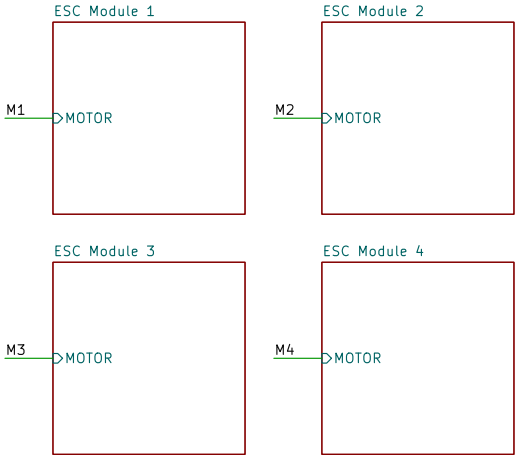
Conns



Power

ESC Modules



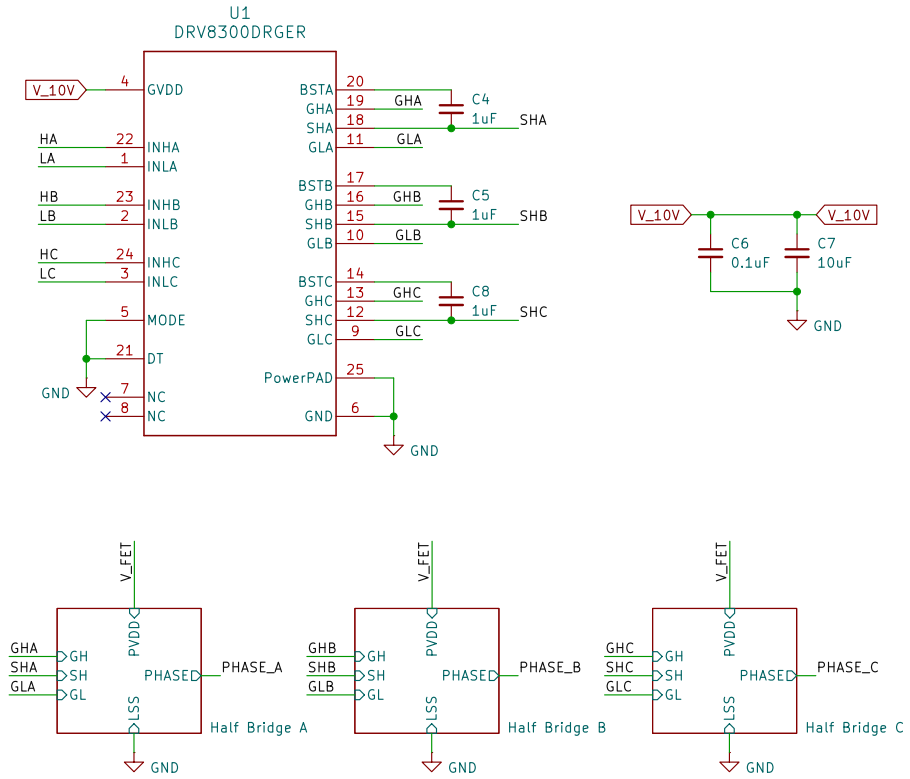
This board placed below FC
~25A continuous per motor, ~100A total continuous
Caps on 10/5V rails: 16V+; VBAT: 30V+
Stratus

Sheet: /
File: esc.kicad_sch

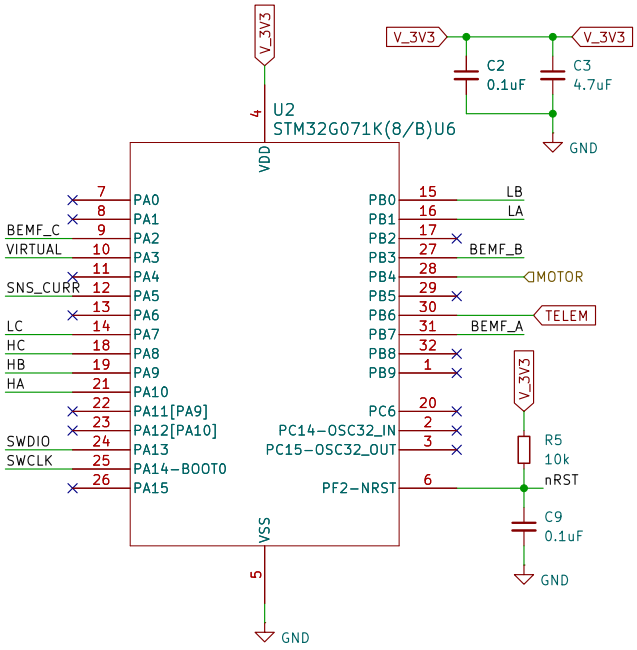
Title: 3–5S 4–in–1 BLDC ESC

Size: A4	Date: 2026–10–13	**Rev: 3**
KiCad E.D.A. 10.0.6		Id: 1/18

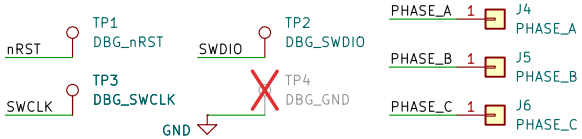
Gate driver + half bridges



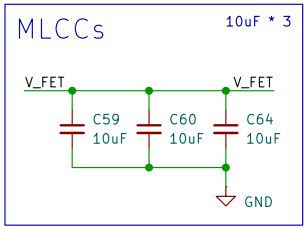
MCU



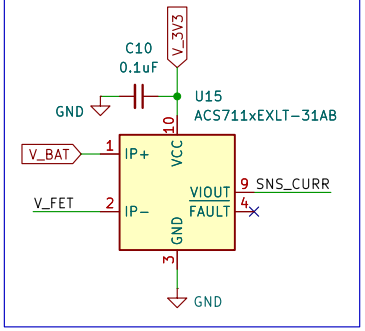
Conns



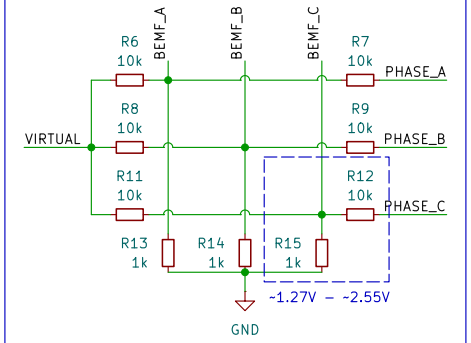
MLCCs



Current sensing



Back-EMF sensing

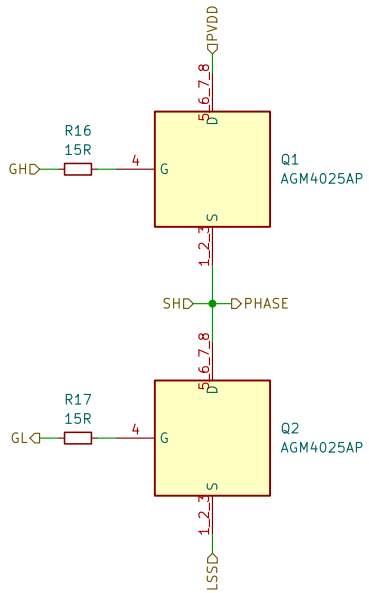


Stratus

Sheet: /ESC Module 1/
File: esc_module.kicad_sch

Title: ESC Submodule

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/18

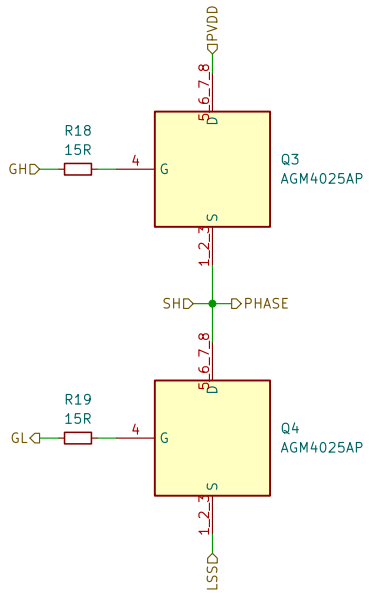


Stratus

Sheet: /ESC Module 1/Half Bridge A/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/18

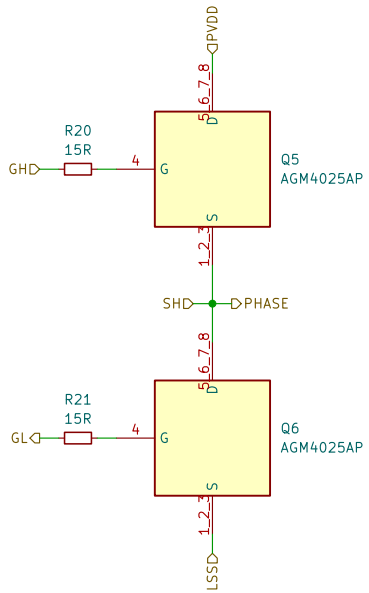


Stratus

Sheet: /ESC Module 1/Half Bridge B/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/18



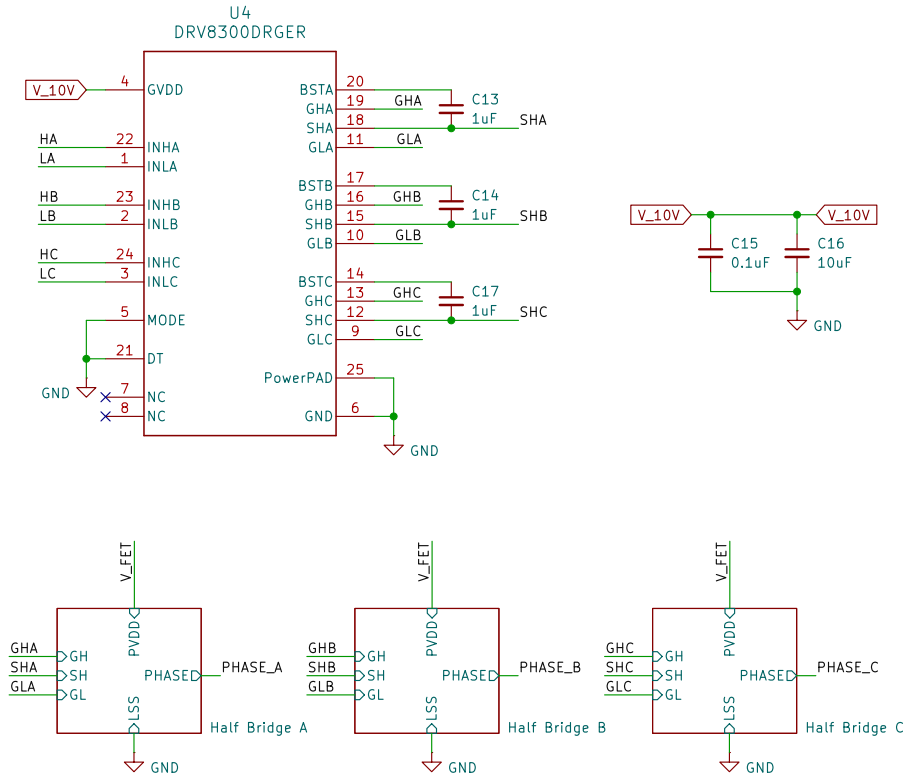
Stratus

Sheet: /ESC Module 1/Half Bridge C/
File: half_bridge.kicad_sch

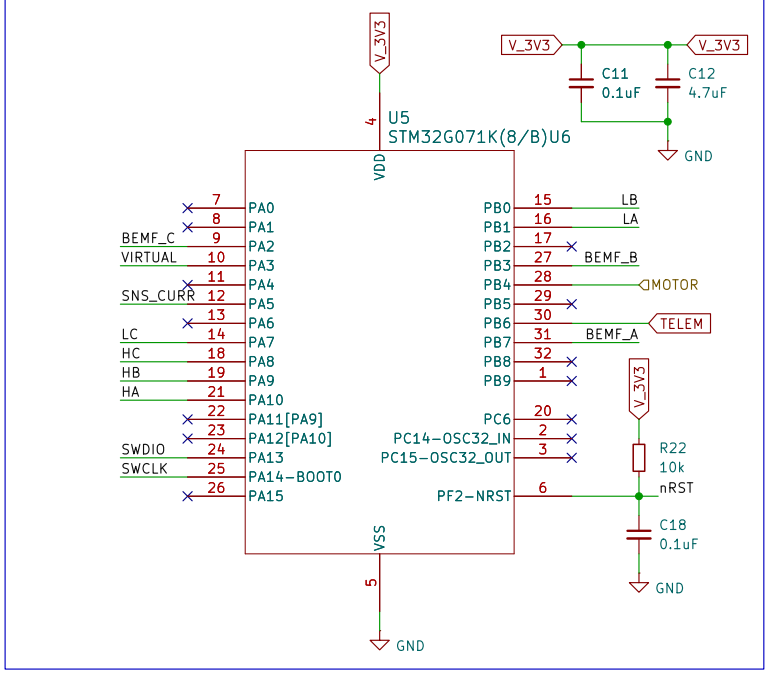
Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/18

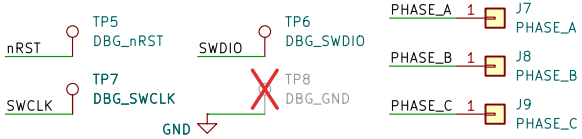
Gate driver + half bridges



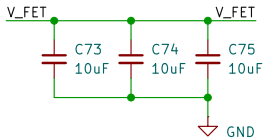
MCU



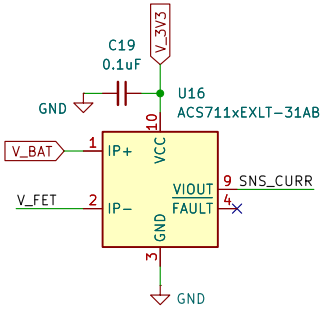
Conns



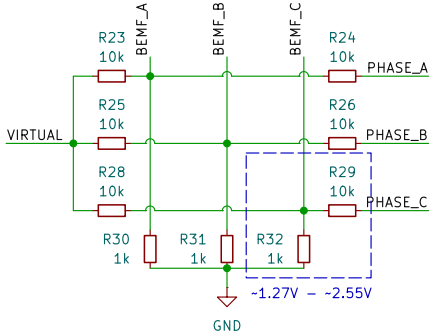
MLCCs



Current sensing



Back-EMF sensing

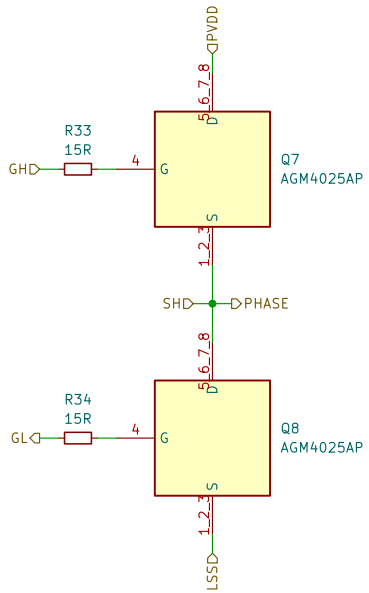


Stratus

Sheet: /ESC Module 2/
File: esc_module.kicad_sch

Title: ESC Submodule

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/18

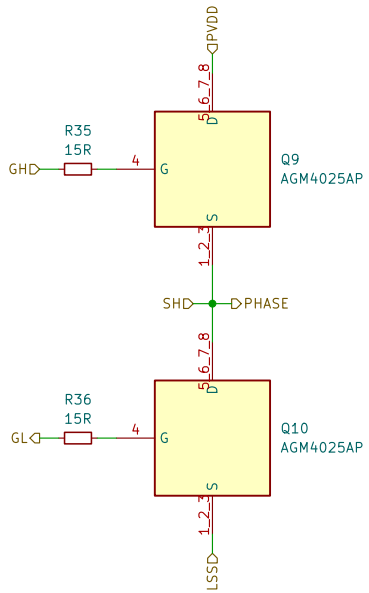


Stratus

Sheet: /ESC Module 2/Half Bridge B/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/18

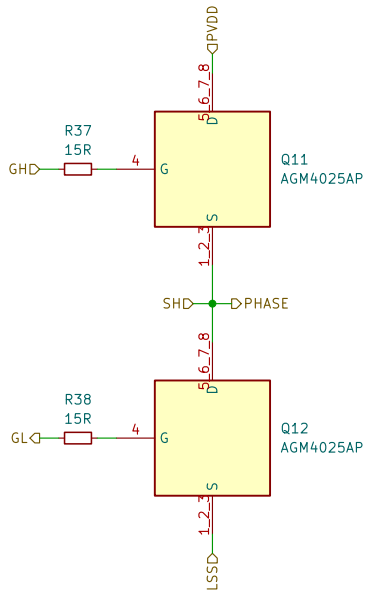


Stratus

Sheet: /ESC Module 2/Half Bridge C/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/18



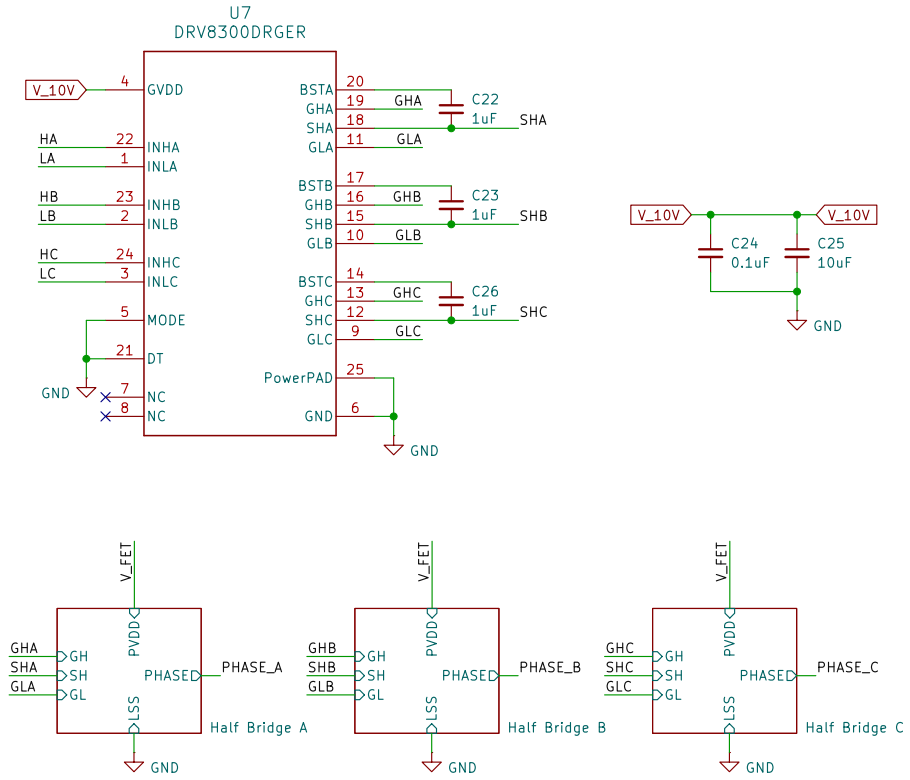
Stratus

Sheet: /ESC Module 2/Half Bridge A/
File: half_bridge.kicad_sch

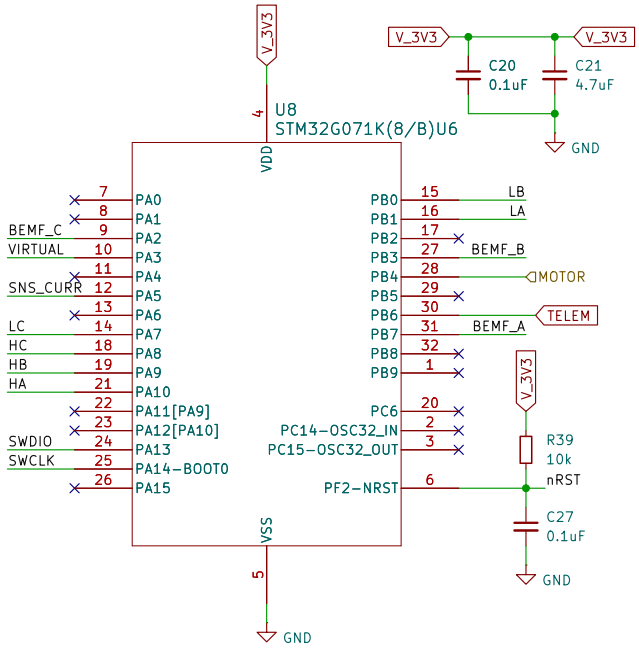
Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 9/18

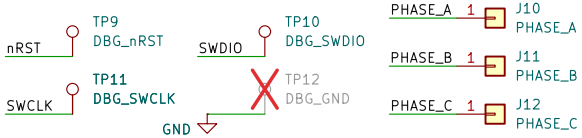
Gate driver + half bridges



MCU

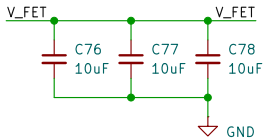


Conns

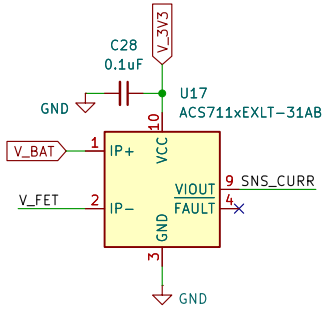


MLCCs

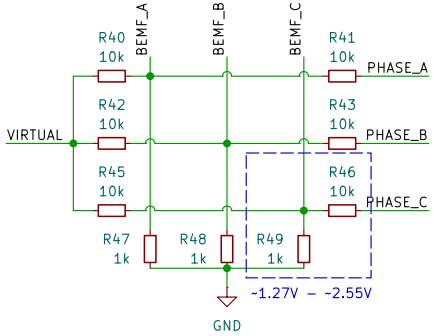
10uF * 3



Current sensing



Back-EMF sensing

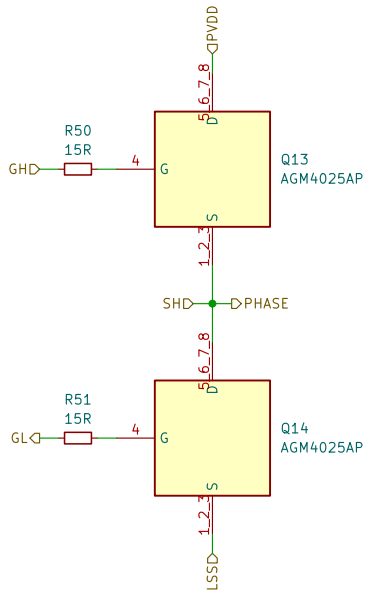


Stratus

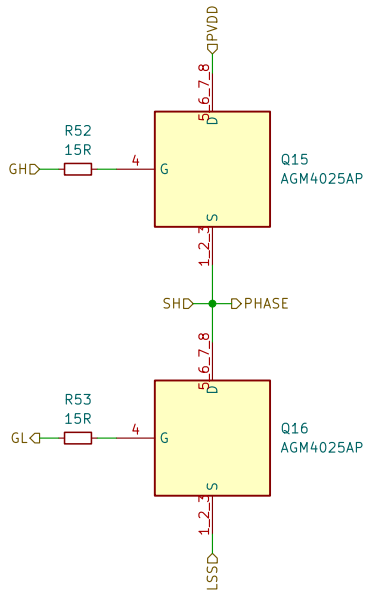
Sheet: /ESC Module 3/
File: esc_module.kicad_sch

Title: ESC Submodule

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 10/18



Stratus		
Sheet: /ESC Module 3/Half Bridge B/		
File: half_bridge.kicad_sch		
Title: MOSFET half bridge		
Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 11/18

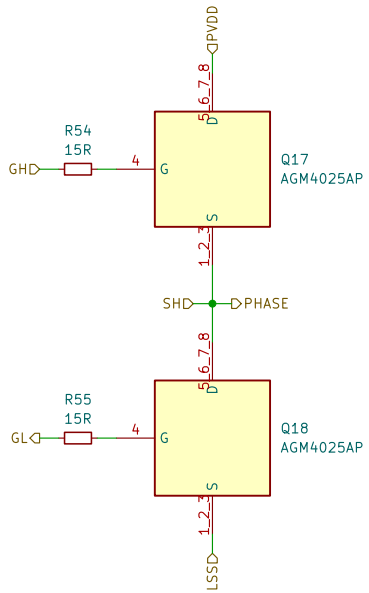


Stratus

Sheet: /ESC Module 3/Half Bridge C/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 12/18



Stratus

Sheet: /ESC Module 3/Half Bridge A/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 13/18

Gate driver + half bridges

U10
DRV8300DRGER

V_10V
GVDD
HA 22 INHA
LA 1 INLA
HB 23 INHB
LB 2 INLB
HC 24 INHC
LC 3 INLC
5 MODE
21 DT
GND
7 NC
8 NC
PowerPAD 25
GND 6 GND

BSTA 20
GHA 19 GHA
SHA 18
GLA 11 GLA
C31 1uF SHA
BSTB 17
GHB 16 GHB
SHB 15
GLB 10 GLB
C32 1uF SHB
BSTC 14
GHC 13 GHC
SHC 12
GLC 9 GLC
C35 1uF SHC

V_10V C33 0.1uF
V_10V C34 10uF
GND

V_FET PVDD
GHA GH
SHA SH
GLA GL
PHASED PHASE_A
LSS
GND
Half Bridge A

V_FET PVDD
GHB GH
SHB SH
GLB GL
PHASED PHASE_B
LSS
GND
Half Bridge B

V_FET PVDD
GHC GH
SHC SH
GLC GL
PHASED PHASE_C
LSS
GND
Half Bridge C

MCU

V_3V3
U11
STM32G071K(8/B)U6
VDD
V_3V3 C29 0.1uF
V_3V3 C30 4.7uF
GND

BEMF_C PA0
VIRTUAL PA1
SNS_CURR PA2
PA3
PA4
LC PA5
HC PA6
HB PA7
HA PA8
PA9
PA10
PA11[PA9]
SWDIO PA12[PA10]
SWCLK PA13
PA14-BOOT0
PA15

PB0 LB
PB1 LA
PB2
PB3 BEMF_B
PB4 MOTOR
PB5
PB6 TELEM
PB7 BEMF_A
PB8
PB9
PC6
PC14-OSC32_IN
PC15-OSC32_OUT
PF2-NRST

VSS
GND

V_3V3 R56 10k nRST
C36 0.1uF
GND

Conns

nRST TP13 DBG_nRST
SWCLK TP15 DBG_SWCLK
SWDIO TP14 DBG_SWDIO
TP16 DBG_GND
GND

PHASE_A 1 J13 PHASE_A
PHASE_B 1 J14 PHASE_B
PHASE_C 1 J15 PHASE_C

MLCCs

10uF * 3

V_FET C79 10uF
V_FET C80 10uF
V_FET C81 10uF
GND

Current sensing

V_3V3 C37 0.1uF
GND
U18
ACS711xEXLT-31AB
V_BAT 1 IP+
V_FET 2 IP-
VCC 10
VIOUT 9 SNS_CURR
FAULT 4
GND 3
GND

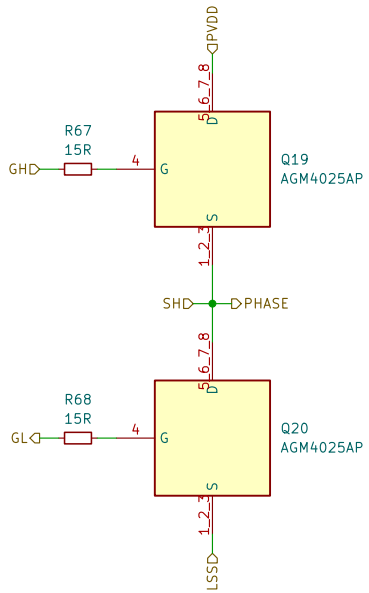
Back-EMF sensing

BEMF_A BEMF_B BEMF_C
VIRTUAL
R57 10k
R58 10k PHASE_A
R59 10k
R60 10k PHASE_B
R62 10k
R63 10k PHASE_C
R64 1k
R65 1k
R66 1k
~1.27V – ~2.55V
GND

Stratus
Sheet: /ESC Module 4/
File: esc_module.kicad_sch

Title: ESC Submodule

Size: A4 | Date: 2026-07-06 | **Rev: A**
KiCad E.D.A. 10.0.6 | Id: 14/18

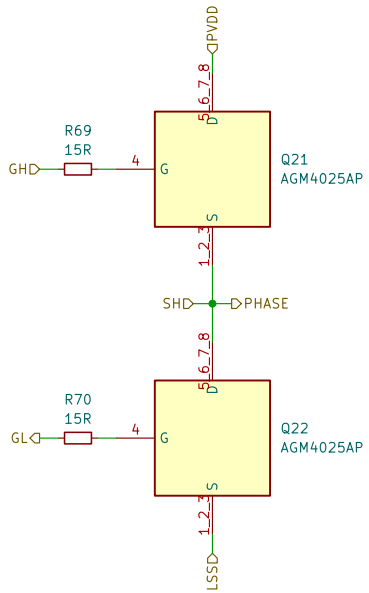


Stratus

Sheet: /ESC Module 4/Half Bridge B/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 15/18

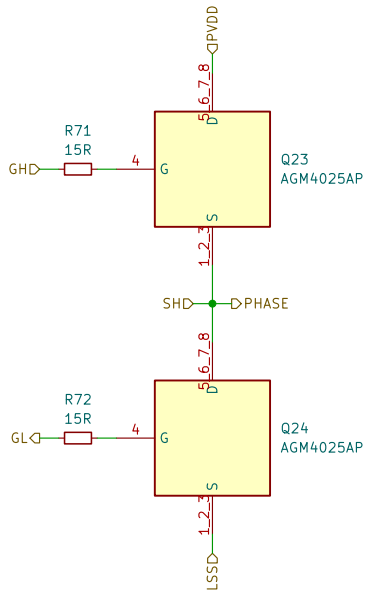


Stratus

Sheet: /ESC Module 4/Half Bridge C/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 16/18



Stratus

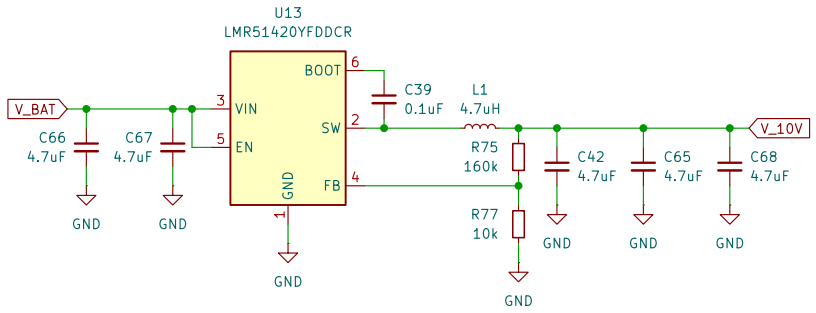
Sheet: /ESC Module 4/Half Bridge A/
File: half_bridge.kicad_sch

Title: MOSFET half bridge

Size: A4	Date: 2026-07-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 17/18

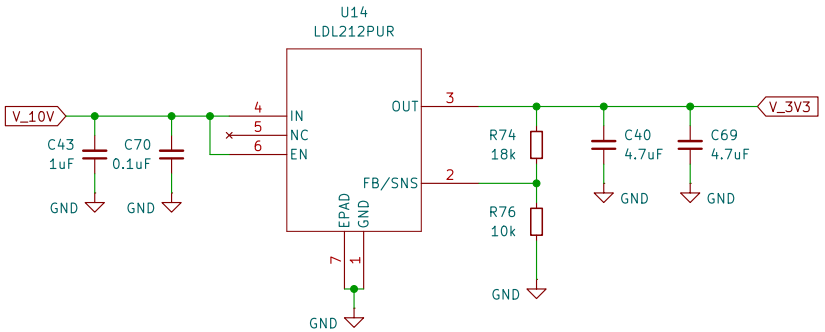
Buck: V_BAT -> 10V

to DRV8300 and LDO



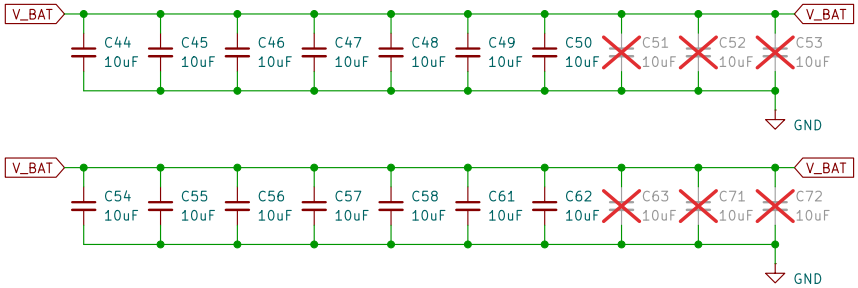
LDO: 10V -> 3V3

to STM32s

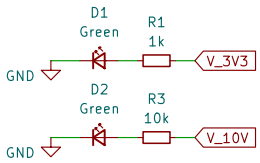


Capacitor bank

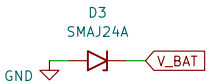
10uF * 12



Indicators



TVS



Stratus		
Sheet: /Power/		
File: power.kicad_sch		
Title: Power Stages		
Size: A4	Date: 2026-09-10	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 18/18